

Underdog Manager

C++ basketball management simulation in Unreal Engine 5.8 with a deterministic match engine

Role: Game systems engineer (C++)

C++

Unreal Engine 5.8

Slate/UMG

Unreal Automation Tests

Architected a C++-first basketball management simulation in Unreal Engine 5.8. You inherit the weakest club in a generated 12-team league and rebuild it over multiple seasons. A deterministic possession-based match engine drives everything, with zero external asset dependencies.

11K+

Lines of C++

21 static service classes, clear module boundaries

22

Dashboard Screens

Built in code with Slate/UMG

47

Automated Tests

Soak, schedule, match, roster, payroll, and cash

Problem

Management sims carry a lot of interlocking state (rosters, finances, staff, morale). Without determinism and tests, one bug can quietly corrupt a ten-season save.

Solution

The core simulation is seed-driven and deterministic, every score and box score is reconciled, saves are versioned and validated, and long soak tests simulate a full career to catch drift.

Architecture

- UE 5.8 runtime and test modules with clear boundaries
- 21 static service classes for league, schedule, match, finances, and staff
- Possession-based match simulator emitting explicit play events
- Game-instance orchestration with atomic round advancement
- Async save snapshots with versioned schema validation
- 22-screen Slate/UMG dashboard and 2D tactical court viewer

Highlights

- Generated 12-team, 180-player league with a validated 22-game double round-robin schedule
- Deterministic possession-based match simulator with overtime and box-score reconciliation
- Standings, playoffs, development, injuries, morale, contracts, free agency, draft, and AI managers
- Async save snapshots with versioned schema validation and deterministic RNG
- 47 passing automated tests, including a ten-season soak
- All UI and audio generated in code, with zero marketplace assets

Challenges and Tradeoffs

- Challenge: Simulation bugs hide over many seasons. Change: A deterministic ten-season soak test covering career, roster, payroll, and cash.

- Challenge: Save formats changed as features grew. Change: A versioned schema validated on load.
- Tradeoff: Licensed art vs zero dependencies. Decision: All UI and audio built procedurally, so the project needs no marketplace content.

Outcomes

- A playable vertical slice covering roster, tactics, trades, playoffs, staff, and finances
- The full automation suite passes headlessly on UE 5.8
- Adds C++ and systems-design depth alongside the web and data work

Repository: <https://github.com/Jaskeerat/underdog-manager> · Full case study: <https://jaskeerat-portfolio.vercel.app/projects/underdog-manager>